VoIP ATA Solutions for the Home and Office

Opening up a new world of IP applications

The business of VoIP in the home and office

Overview

DSP Group™'s comprehensive VoIP solution portfolio provides residential and business customers with the benefits of low-cost analog terminal adapter (ATA) applications, as well as easy and natural expandability to cordless telephony.

DSP Group's software development kit (SDK) and accompanying reference materials help equipment manufacturers build cost-effective, multi-line voice ATA applications with easy extension to DECT for supporting multi-handsets and integrated mini-PBXs with HD voice.

Features

- Highly-integrated digital and analog IC solution for system bill of materials (BOM) cost savings and compact design
- Up to 6 FXS and/or FXO ports
- Simultaneous VoIP call support (up to 6 calls)
- G.729a, G.723.1, G.711, G.711.1 WB, iLBC, G.722, G.722.2 (WB-AMR), G.726 vocoders
- SIP stack and SIP application (RadVision or others)
- Fax data pump (V.21, V.27, V.29, V.17) and T.38 stack
- Integrated PBX functions
- Superb voice quality including echo cancellation, comfort noise generation (CNG), voice activity detection (VAD), packet loss concealment (PLC), automatic gain control (AGC), flexible noise reduction (FNR), and adaptive jitter buffer
- IP protocol support including IP/TCP/UDP/ICMP/ARP/RARP, DHCP client, PPPoE, TFTP, HTTP, SNTP & DNS & FTP, SSH
- Web management including TR.69 engine
- Strong NAT routing support, TCP bandwidth up to 70 Mbps
- RTP and RTCP applications
- Various ptimes (up to 90 ms) and on-the-fly codec change support
- QoS (IEEE 802.1P, IEEE 802.1Q & ToS field)
- RFC 3261, 3263, 3264, 3265, 4733, 2833, 3550, 3551, 4028, 2976, 3311, 3323, 4585, 5505, and more
- Configuration including web-based GUI, menu on the device, and auto provisioning system (HTTP/TFTP)
- Integrated Skype engine
- Integrated digital answering machine (DTAM) (optional)
- USB device support including serial, Ethernet and audio gadgets
- USB host support with external controller
- SMS over IP and PSTN
- Security functions including AES engine, SRTP
- Voice statistics support and loop back for on-field monitoring and debugging
- Software upgradability
- Low power consumption <0.5W at idle

Seamless DECT Extension

- Integrated RF IC including advanced DECT transceiver, PA and high-performance LNA with superior sensitivity
- Worldwide DECT standard support
- CAT-iq and GAP compliance
- HD voice support, AEC, polyphonic ringing
- Multi-handset (up to 10 active) support with multi-line PBX features
- Software upgradability over the air (SUOTA along CAT-iq 3.0) and data services over the air

Benefits

- Low cost and high performance for basic ATAs and up
- Strong standard processors (ARM9™ running Linux OS and high-performance DSP)
- Full set of hardware and software reference designs, SDKs and single-source applications enabling short development cycle
- Easy extendibility to CAT-iq connectivity

Applications

- Fully featured analog telephony adapter (ATA)
- Cordless ATA with DECT handsets and/or headsets
- Home PBX access point
- VoIP gateway
- Embedded Skype
- USB dongle with telephony extension (with/without DECT)

DSP GROUP

at home in your wireless world

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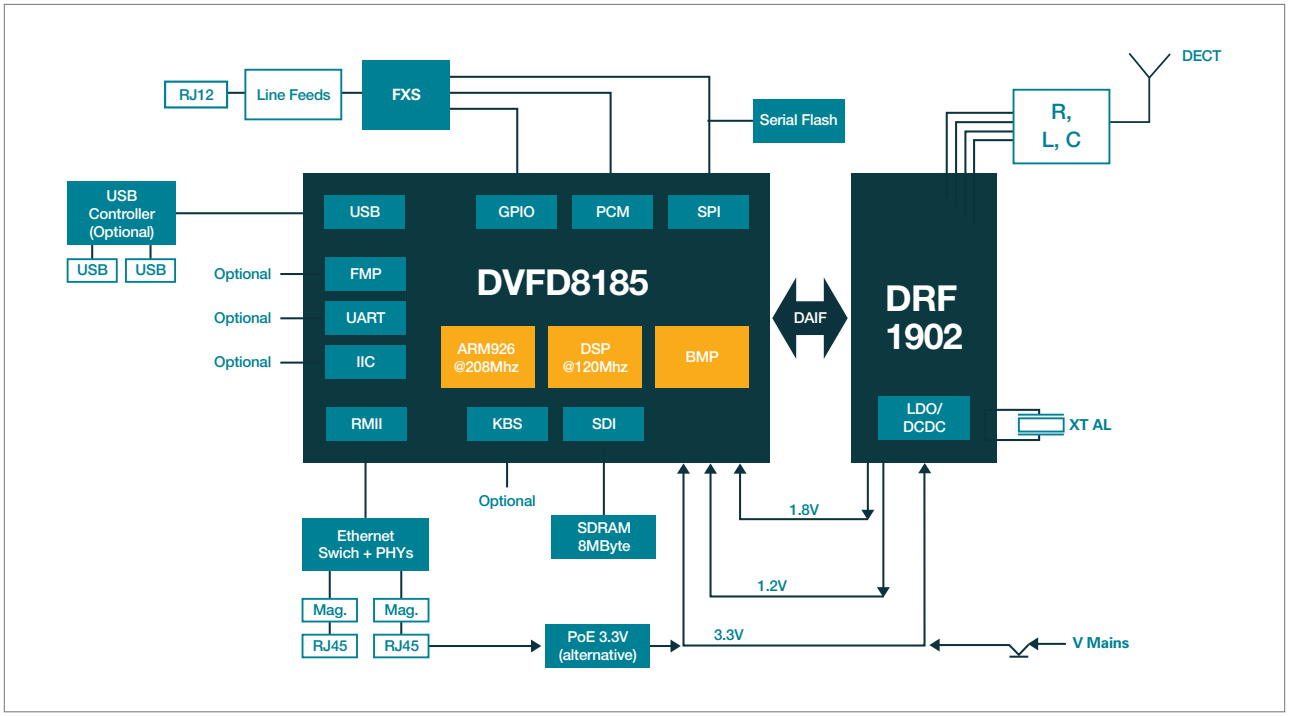
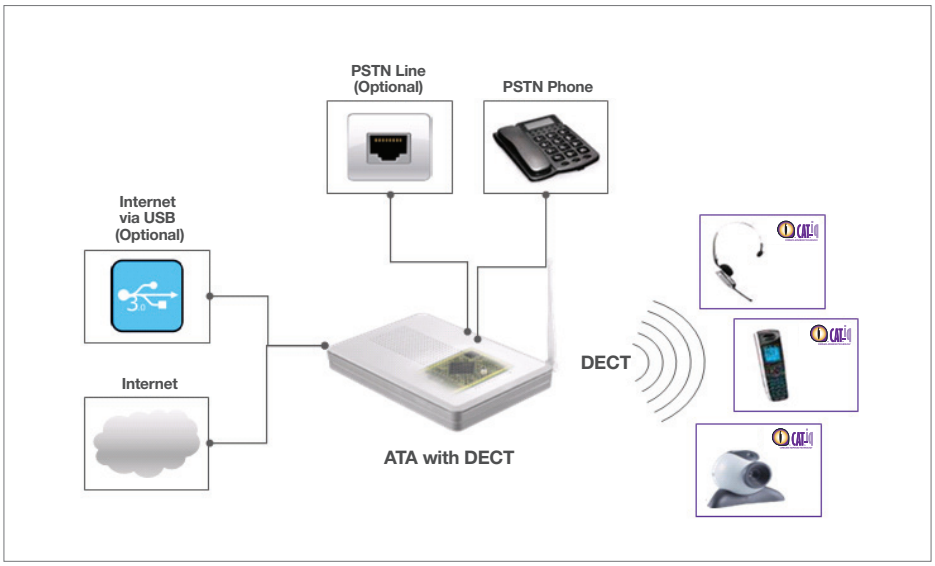
Block Diagram



System Connectivity



DSP Group, Inc. (NASDAQ: DSPG) is a leading global provider of wireless chipset solutions for converged communications at home. Delivering system solutions that combine semiconductors and software with reference designs, DSP Group enables consumer electronics (CE) manufacturers to cost-effectively develop new revenue-generating applications with fast time to market. At the forefront of semiconductor innovation and operational excellence for over two decades, and with a growing share of the wireless home telephony market, DSP Group provides a broad portfolio of wireless chipsets integrating DECT, Wi-Fi, PSTN and VoIP technologies with state-of-the-art application processors. Enabling converged voice, audio, video and data connectivity across diverse consumer products – from cordless and VoIP phones to home gateways and connected multimedia screens – DSP Group proactively partners with CE manufacturers to shape the future of converged communications at home.

For more information, visit **www.dspg.com**

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